

Proposal: Research Center for Quantum Control

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1 Introduction and Vision

Quantum technologies such as quantum computing, simulation, and sensing have advanced rapidly in recent years, yet their full potential can only be realized through precise control of quantum systems. Active control of quantum dynamics — the ability to shape electromagnetic pulses and external fields to steer quantum states — is a foundational capability enabling high-fidelity operations in complex quantum devices. Numerous studies have shown that it is often surprisingly feasible to locate optimal controls for even complex quantum phenomena, owing to favorable properties of quantum control landscapes [?]. For example, unlike many classical systems, quantum systems tend to lack ubiquitous false optima in their control landscapes, meaning that gradient-based algorithms can often efficiently discover controls that drive a system to a desired target state or operation. This insight, developed through theoretical analyses of control landscapes [?], underpins the optimistic vision that quantum hardware with many coupled degrees of freedom can be tamed via systematic control design.

Building on these theoretical foundations, the proposed Research Center for Quantum Control will unite leading experts in quantum control theory and experiment to overcome current limits in our ability to manipulate quantum systems. The Center’s vision is to establish a world-class, cross-disciplinary hub that integrates fundamental theory with state-of-the-art experimental platforms, ranging from superconducting qubits and trapped ions to neutral atoms and photonic systems. By tightly coupling theoretical advances (e.g., in coherent control of quantum dynamics, control of open quantum systems under decoherence, and optimal control algorithms) with experimental implementation on real quantum hardware, we aim to dramatically improve the performance, scalability, and reliability of quantum technologies. This includes pushing the frontier of high-fidelity quantum gates and error-resilient operations in quantum computers, unlocking new regimes of quantum simulations through fine-tuned Hamiltonian engineering, and enhancing quantum sensors and metrological devices via optimal quantum state preparation and control.

Another key element of our vision is to create shared infrastructure and collaborative environments that enable rapid testing of control techniques across multiple platforms. The Center will operate dedicated quantum hardware testbeds where novel control schemes (for example, new pulse sequences or real-time feedback protocols) can be evaluated in superconducting circuit experiments, ion trap systems, atomic arrays, and photonic networks. This comparative approach will allow us to identify common principles and unique challenges in controlling different physical systems, fostering the development of universal control strategies as well as platform-specific solutions. Equally important is the Center’s mission to train the next generation of quantum scientists and engineers in the principles and practice of quantum control. Through integrated research and education programs, students and postdoctoral researchers will gain hands-on experience with cutting-edge control software and hardware, preparing them to become future leaders in academia, industry, and national labs.

In summary, the Research Center for Quantum Control will serve as a focal point for advancing the science and technology of quantum control. By bringing together theory, experimental platforms, and com-

putational tools under one umbrella, the Center will accelerate progress toward error-corrected quantum computation, high-precision quantum measurements, and other transformative applications of controlled quantum systems. The following sections detail our scientific objectives, research program, required infrastructure, educational plans, management structure, and the broad impacts expected from this initiative.

2 Scientific Objectives

The Center’s scientific agenda is organized around several key objectives that address both fundamental research questions and technology development needs in quantum control. These objectives, which collectively define our research scope and priorities, are as follows:

1. **Advance Fundamental Quantum Control Theory:** Develop deeper theoretical understanding of quantum control mechanisms, including coherent control in closed systems, control of open quantum systems with dissipation, and the topology of quantum control landscapes. This objective includes elucidating why quantum control landscapes often lack suboptimal traps and how quantum interference can be harnessed to steer dynamics, thereby guiding the design of more effective control algorithms [?, ?].
2. **Innovate Optimal Control Methods and Algorithms:** Create and refine computational methods for optimal control of quantum processes. This includes improving gradient-based pulse shaping techniques (e.g., GRAPE) and other numerical algorithms to discover control pulses that achieve desired quantum gate operations or state transformations with high fidelity [?]. We will also explore new approaches like machine learning optimization for quantum control, and develop open-source software tools that implement these algorithms for broad use by the community.
3. **Integrate Theory with Experimental Quantum Platforms:** Demonstrate advanced control techniques on leading physical hardware platforms — superconducting qubits, trapped ions, neutral atoms, and photonic quantum systems. For each platform, our objective is to push the state-of-the-art in control performance (e.g., higher gate fidelities, longer coherence under active error suppression, more precise quantum state preparation) by applying optimized pulses, real-time feedback, and other control innovations. We aim to close the gap between theoretical control designs and laboratory implementation, validating theory on hardware and informing new theory from experimental discoveries.
4. **Develop Robust Error Mitigation and Noise-Management Strategies:** Create control-based solutions to mitigate the effects of decoherence and operational imperfections in quantum devices. This objective includes designing dynamical decoupling sequences and error-resistant control protocols that can extend coherence times [?], implementing measurement-driven feedback or feed-forward control to stabilize quantum states, and integrating control with emerging error mitigation techniques for NISQ (Noisy Intermediate-Scale Quantum) processors [?]. Our goal is to significantly reduce logical error rates in quantum computations and improve the reliability of quantum sensors by active control of noise.
5. **Enable Applications in Computing, Simulation, and Metrology:** Apply the advances in quantum control to realize cutting-edge demonstrations in quantum computing (e.g., high-fidelity multi-qubit quantum gates and improved quantum algorithms on controlled hardware), quantum simulation (e.g., precisely engineered Hamiltonian dynamics in many-body systems for modeling complex quantum phenomena), and quantum metrology (e.g., preparation of entangled sensor states that surpass classical measurement limits [?]). By meeting this objective, the Center will show how improved control directly translates to enhanced capabilities in practical quantum applications, such as achieving quantum advantage in computation or reaching unprecedented measurement precision.

These five objectives are interlocking and will be pursued in parallel by different teams within the Center, with strong communication and feedback among them. The fundamental theory developments (Objective 1) will feed into improved algorithms and tools (Objective 2), which in turn drive experimental implementations (Objective 3). Progress in error mitigation (Objective 4) will be essential for success in all hardware demonstrations and applications (Objective 5). Together, these objectives form a comprehensive strategy to elevate quantum control from basic theory through practical technology and ultimately to societally relevant quantum systems.

3 Research Program

To achieve the above objectives, the Center’s research program is structured into several thrusts that integrate theoretical and experimental efforts across the various quantum platforms. Each thrust is a collaborative endeavor led by experts in the respective subfield, ensuring cross-fertilization of ideas and techniques. The major research thrusts are:

- **Quantum Control Theory and Algorithms:** This thrust focuses on advancing the underlying theory of quantum control and developing new optimal control algorithms. Researchers will investigate coherent control techniques (e.g., shaping of pulses to exploit quantum interference) and control of open quantum systems via noise filtration and feedback. They will analyze control landscapes to understand the ease of finding optimal solutions [?] and design globally efficient algorithms for pulse optimization. Novel algorithms (such as gradient ascent pulse engineering (GRAPE) and Krotov methods) will be refined for faster convergence and incorporated into software tools [?]. This theoretical work provides the foundation that guides all experimental thrusts.
- **Superconducting Qubit Control:** Superconducting quantum circuits are one of the flagship platforms for quantum computing. In this thrust, we develop advanced control techniques for superconducting qubits (e.g., transmons and flux qubits) to achieve error rates well below the threshold for quantum error correction. The team will implement optimized microwave pulse shapes to realize high-fidelity single- and two-qubit gates, as well as tunable couplings for multi-qubit operations. Recent progress has shown that multiple high-fidelity two-qubit gates (with error probabilities below 1%) and even operations on logical qubits in extensible superconducting systems are now possible [?]. Building on this, we will explore control strategies such as DRAG (Derivative Removal by Adiabatic Gate) pulses to minimize leakage, flux modulation schemes for fast two-qubit entangling gates, and real-time calibration routines. The superconducting qubit testbed will also serve as a platform for demonstrating feedback control and error mitigation techniques (like dynamical decoupling integrated with fast control electronics) to prolong coherence. Our aim is to validate that improved control can enable complex algorithms on superconducting processors with significantly higher fidelity than current state-of-the-art.
- **Trapped Ion Control:** Trapped ions offer superb coherence properties and high-fidelity gate operations using laser or microwave-driven interactions. Here, the research will focus on scaling up control as the number of ion qubits grows, and on speeding up gate operations without sacrificing fidelity. The team will develop pulse sequences and scheduling techniques to mitigate crosstalk and motional mode interference in multi-ion chains. All basic requirements for universal quantum computing have been demonstrated with few-ion systems [?], and we will build on this by using optimal control (e.g., amplitude and frequency modulation of laser pulses) to implement faster entangling gates (such as Mølmer-Sørensen gates) with minimal error. We will also investigate adaptive laser pulse shaping and real-time feedback from ion state detection to correct small deviations on-the-fly. A trapped-ion

experimental setup in the Center will serve as a testbed for these methods, aiming to maintain ion qubit coherence and gate fidelity as systems are scaled to tens of qubits. Control solutions developed here, such as multi-tone pulses or dynamically optimized decoupling sequences tailored for ion motion, will inform techniques in other platforms as well.

- **Neutral Atom and Rydberg Control:** Neutral atom arrays (optical tweezers trapping individual atoms) with Rydberg-mediated interactions have emerged as a powerful platform for quantum simulation and computing. The Center will pursue control techniques to harness these systems' intrinsic scalability to hundreds of qubits [?]. One focus is on precise spatial and temporal control of the optical tweezers and laser excitation pulses to achieve consistent two-atom entanglement across large arrays. By shaping the laser pulses driving Rydberg transitions, we aim to optimize gate fidelities and reduce sensitivity to atomic motion or laser frequency noise. The level of single-particle control already achieved (manipulating individual atoms while preserving coherence and entanglement) makes this platform very promising [?]. We will extend that control to multi-atom operations by implementing protocols for parallel gate execution and crosstalk avoidance. Additionally, this thrust will explore controlling atomic internal states for quantum error correction codes (e.g., dynamically reconfiguring tweezer patterns to perform logical operations). Collaborations between theorists and experimentalists will yield algorithms for pulse schedules that realize effective Hamiltonians for quantum simulation tasks, such as realizing spin models or preparing entangled states for metrology.
- **Photonic Quantum Control:** Photonic systems (single photons in optical circuits or nonlinear optics setups) require a different approach to control, as photons do not naturally interact with one another and propagate at the speed of light. Here, control focuses on the generation, routing, and measurement of photonic qubits with high fidelity. The research team will work on active photonic circuit elements such as electro-optic modulators and dynamically reconfigurable interferometers to control photonic qubit modes. One goal is to implement real-time feed-forward protocols in photonic quantum computing, where the outcome of one measurement is used to adjust subsequent photon operations, effectively closing a control loop at high speed. Another focus is on controlling sources of single or entangled photons (e.g., quantum dot emitters or parametric down-conversion sources) to deliver photons on-demand with desired temporal shapes. Advances in this area leverage the unique advantages of photonics, such as low decoherence and inherent compatibility with communication. Recent developments highlight photonic architectures as a promising approach for scaling quantum computers in the NISQ era and beyond [?]. Our program will integrate such architectures with control techniques, for example by using optimized gating and synchronization of photons to implement deterministic multi-photon logic. Though photonic qubits are less susceptible to environmental decoherence, control is key in synchronizing and interfering photons for computation and in mitigating loss errors through redundancy or heralding techniques.
- **Quantum Control Software and Pulse Engineering:** A dedicated effort in the Center will concentrate on developing software tools, simulation frameworks, and firmware for quantum control. We will create a unified control software stack that allows researchers to design pulse sequences and control protocols in simulation and then deploy them on real hardware seamlessly. This includes extending open-source frameworks (such as IBM's Qiskit Pulse) to support multiple hardware backends and incorporating our optimal control algorithms into user-friendly libraries. The Center will also design custom electronics and FPGA-based controllers capable of generating complex pulse shapes with precise timing, which are crucial for implementing the optimized control solutions discovered by our algorithms. By providing an integrated set of control tools and a library of well-tested pulse sequences (for example, error-robust composite pulses and dynamically corrected gates), we aim to accelerate the adoption of advanced control methods in quantum experiments. This thrust ensures that the theo-

retical and experimental progress is translated into accessible technology that can be disseminated to other labs and even industrial partners.

- **Optimal Control and Error Mitigation Strategies:** In conjunction with the platform-specific work, a cross-cutting thrust will focus on error mitigation through control. Techniques like dynamical decoupling, which apply sequences of pulses to average out environmental noise, will be optimized for each system [?]. We will investigate pulse schedules that suppress specific noise spectra (e.g., $1/f$ flux noise in superconducting qubits or magnetic field noise in trapped ions) and experimentally validate the extended coherence. Moreover, the team will combine control with emerging error mitigation protocols such as probabilistic error cancellation and zero-noise extrapolation in quantum computations [?]. By integrating these methods into the control layer (for instance, adjusting pulses adaptively based on calibration data to effectively cancel errors), we expect to bridge the gap between noisy hardware and the requirements of fault-tolerant algorithms. The outcome will be a suite of control-based error reduction techniques that can be applied without the full overhead of quantum error correction, offering immediate improvements in NISQ devices.

The research program is highly collaborative: the thrusts described above will not operate in isolation, but rather exchange techniques and insights. For example, a pulse shaping method developed in the superconducting qubit context may inspire improved laser control in the trapped ion system; conversely, an error mitigation approach from the ion trap (such as a particular dynamical decoupling sequence) could be translated to superconducting circuits. Regular cross-thrust meetings and an internal repository of control protocols will facilitate this knowledge transfer. The Center will also maintain an experimental “exchange” program where students and postdocs spend time in different platform labs to learn and port control techniques between them. Through these mechanisms, the diverse research activities will reinforce each other, driving the entire field of quantum control forward.

4 Infrastructure and Facilities

Achieving the Center’s goals requires specialized infrastructure and shared facilities that provide state-of-the-art experimental and computational capabilities. The following resources will be established and made accessible to all Center participants:

- **Superconducting Quantum Hardware Facility:** A laboratory equipped with multiple dilution refrigerators capable of cooling superconducting qubit chips to millikelvin temperatures. This facility will house superconducting qubit devices (transmon qubit processors and custom test circuits), along with high-speed microwave control electronics and room-temperature infrastructure for signal generation and readout. It will enable experiments on multi-qubit superconducting processors for testing control sequences and error mitigation in a controlled environment.
- **Trapped Ion Quantum Laboratory:** A facility with ultra-high vacuum chambers and ion trapping systems (radio-frequency Paul traps and/or Penning traps) to host chains of trapped ion qubits. It will include stable laser systems for cooling and quantum gate operations (with high-bandwidth modulators for pulse shaping), high-numerical-aperture optics for qubit state detection, and real-time classical controllers. This lab allows the implementation and evaluation of advanced laser pulse sequences and feedback control on trapped ions in a high-coherence setting.
- **Neutral Atom and Optical Lattice Facility:** A laboratory containing apparatus for neutral atom quantum computing and simulation, including optical tweezer arrays and optical lattice setups. High-power, finely tunable lasers for cooling, trapping, and exciting Rydberg states will be available, along

with spatial light modulators and beam steering systems to dynamically reconfigure atomic arrays. Fast single-photon detectors and cameras will support measurement and feedback. Researchers will use this facility to apply control schemes for large atom arrays and to explore scalable quantum simulation under precise control.

- **Photonic Quantum Systems Lab:** A dedicated space for photonic quantum information experiments, featuring sources of single photons (quantum dot emitters and spontaneous parametric down-conversion setups) and reconfigurable photonic circuits (fiber-based and integrated waveguide platforms). The lab will be outfitted with high-speed optical modulators, switches, and timing electronics needed for feed-forward logic with photons. This facility will enable testing of photonic control protocols such as real-time adaptive interferometry and synchronization of probabilistic photon processes for computation.
- **Shared Control Electronics and Integration:** A centralized electronics workshop and server hub that provides critical infrastructure for all experimental thrusts. This includes custom arbitrary waveform generators, FPGA-based real-time control systems, and data acquisition units that can be interfaced with each quantum experiment. A networked control framework will allow experiments to be monitored and operated remotely, facilitating multi-institution collaborations and external user access to the testbeds. By standardizing on a common control hardware/software stack (with appropriate adapters for each platform), the Center will simplify the deployment of new control techniques across different experimental setups.
- **Computational Resources:** A high-performance computing cluster dedicated to quantum control simulations and optimization tasks. Equipped with both CPUs and GPUs, this cluster will run intensive optimal control algorithms (for example, solving Schrödinger or master equations iteratively for pulse optimization) and machine learning training for control policies. It will also host the Center's repositories for simulation software, pulse sequence libraries, and experimental data. The computational facility ensures that theoretical predictions and designs can be rapidly generated and iteratively refined using feedback from experimental results.
- **Nanofabrication and Materials Support:** Through partnerships with the university's nanofabrication center, the program will have access to cleanroom facilities for fabricating or customizing quantum devices (such as specialized qubit chips or photonic circuits) that incorporate features to facilitate control (e.g., on-chip filters, custom wiring for control lines, etc.). Additionally, materials characterization tools (for superconducting film quality, optics coatings, etc.) will support the development of robust hardware for control experiments.

All these facilities will be operated as shared user facilities within the Center, promoting a culture of open access and collaboration. A centralized scheduling and support system will allow researchers (including students and external collaborators) to request time on the different experimental testbeds. Technical staff, including laboratory engineers and technicians, will maintain the equipment, assist with experiment integration, and ensure that new control apparatus (like novel waveform generators or laser systems) can be reliably used across multiple projects. This shared infrastructure approach avoids duplication of costly equipment and creates an environment where theoretical ideas can be promptly tested on real quantum devices. It is a critical enabler for the Center's interdisciplinary research, as it lowers the barrier for any researcher to work with any platform using the best available tools.

5 Education and Outreach

A core mission of the Center is to cultivate a highly skilled quantum workforce and to engage the broader community through outreach. Our education and outreach plans include:

- **Curriculum and Coursework:** Development of new academic courses and training programs focused on quantum control and quantum engineering. The Center faculty will create graduate-level courses covering topics like quantum control theory, optimal control algorithms, quantum hardware programming (e.g., pulse-level control in quantum computing), and error correction/mitigation techniques. We will also introduce hands-on laboratory courses where students can practice implementing control experiments on simplified setups (such as NMR analogs for spin control or photonics kits for single-photon experiments). These courses will be made available to students across physics, electrical engineering, and computer science departments, reflecting the interdisciplinary nature of quantum control.
- **Student and Postdoctoral Training:** The Center will support a cohort of graduate students and postdocs working on its research projects, providing them with unique cross-disciplinary mentorship. Each young researcher will have both a theoretical mentor and an experimental mentor, ensuring they gain experience in both domains. Through rotations or joint projects, a student might, for example, work on pulse optimization theory and then spend time in the lab applying those pulses on a quantum device. Such integrated training will produce researchers who are fluent in both the abstract and practical aspects of quantum control. The Center will also host regular internal seminars, tutorial sessions, and an annual retreat where students and postdocs present their work and network with invited experts.
- **Workshops and Summer Schools:** To disseminate knowledge beyond our institutions, we will organize an annual Quantum Control Workshop open to participants from other universities, national labs, and industry. This workshop will feature tutorials on the latest control techniques and hands-on sessions using our open-source software and remote access to our testbed devices. Additionally, we plan to run a week-long Summer School on Quantum Control each year for advanced undergraduates and beginning graduate students. Topics will range from fundamentals of coherent control to practical laboratory implementation, aiming to inspire and prepare the next generation of researchers in this field.
- **Outreach and Broadening Participation:** The Center is committed to broadening participation in quantum science. We will engage in outreach to local schools and the public, highlighting the excitement of quantum technology and the role of control in enabling it. Planned activities include demonstrations of quantum control principles using simple analogs (e.g., using polarized light or two-level electronic systems to illustrate qubit control), public lectures or webinars by Center members, and development of educational multimedia content (such as short videos explaining how quantum devices are controlled). We will actively recruit students from underrepresented groups through partnerships with minority-serving institutions and through programs such as summer research internships for undergraduates. By fostering an inclusive environment and showcasing diverse role models among our researchers, we aim to contribute to a more diverse quantum science community.
- **Industry and Public Sector Engagement:** Recognizing the growing quantum industry, the Center will maintain ties with companies and government labs interested in quantum control. We will invite industry experts for guest lectures and collaborate on applied research projects (when appropriate) to transition control techniques into real-world quantum technology deployments. Our outreach will also involve communicating our findings to policy makers and the general public to raise awareness

of quantum technology progress and the need for continued research investment. By training students who understand both fundamental science and practical implementation, we will create a talent pipeline that benefits the emerging quantum economy.

Through these education and outreach initiatives, the Center will not only advance research but also ensure that its impact is amplified via human capital development and public engagement. The combination of specialized training for students, knowledge-sharing events for the broader community, and deliberate efforts to include diverse participants will help build a robust and inclusive field of quantum control that extends well beyond the lifetime of this grant.

6 Management Plan

The Center for Quantum Control will be managed via a clear organizational structure and proactive leadership to coordinate its interdisciplinary activities. The management plan covers governance, internal communication, external advisory input, and evaluation of progress.

Leadership and Governance: The Center will be led by a Director (the Principal Investigator of the grant) who is responsible for overall strategic direction, resource allocation, and ensuring that the Center’s objectives are met. The Director will be supported by co-Principal Investigators, each of whom coordinates one of the major research thrusts (Theory, Superconducting, Ion, Atomic, Photonic, etc.) or cross-cutting areas (Education/Outreach, Software/Infrastructure). Together, the Director and co-PIs form an Executive Committee that meets weekly to discuss progress, troubleshoot issues, and plan upcoming activities. The Center will also establish a Steering Committee composed of all thrust leaders and key institutional partners, meeting monthly to provide broader input into management decisions and to promote integration across thrusts.

Collaboration and Communication: Given the multi-institutional nature of the Center (involving multiple departments and universities), effective communication channels will be crucial. We will implement regular all-hands meetings (e.g., a video conference every two weeks) where researchers give brief updates, share interim results, and discuss any challenges or opportunities. An online collaboration platform will be used to share documents, software code, experimental data, and schedules for the shared facilities. Within each research thrust, sub-teams will have more frequent meetings, but we will also form cross-thrust working groups (for instance, a working group on “optimal control software” including both theorists and experimentalists from different platforms) to tackle specific interdisciplinary problems. The management will encourage personnel exchanges and lab visits to strengthen inter-group collaboration and knowledge transfer.

Advisory Board: We will convene an External Advisory Board consisting of renowned experts in quantum control, quantum information technology, and related fields (including members from industry and national research organizations). The Advisory Board will meet annually to review the Center’s progress, provide feedback on research directions, and offer guidance on maximizing impact. Prior to each annual meeting, the Center will prepare a comprehensive report of its research results, educational activities, and expenditures, which the Advisory Board will evaluate. The Board’s recommendations will be documented and used by the Executive Committee to refine the Center’s strategic plan and to address any identified weaknesses or new opportunities. This external input will ensure that the Center remains aligned with national priorities and aware of emerging developments in the field.

Evaluation and Milestones: The Center’s progress will be tracked against a set of milestones defined for each objective and research thrust. For example, milestones may include achieving a target gate fidelity on a specific platform, releasing a version of a control software toolkit, or demonstrating a particular reduction in error rates via control. The Executive Committee will conduct internal evaluations every six months,

assessing which milestones have been met and identifying any areas that require corrective action or additional support. If certain research directions are not yielding the expected results, the leadership will be prepared to pivot strategies or reallocate resources (in consultation with the relevant teams) to better achieve the Center’s goals. We will also collect quantitative metrics such as number of publications, software downloads, workshop participants, and students trained, to measure the impact of the Center’s activities over time.

Administrative Support and Reporting: The Center will employ dedicated administrative staff (e.g., a program manager and an administrative assistant) to handle day-to-day operations, coordination, and reporting. The program manager will assist the Director in budget management, ensure compliance with grant requirements, organize meetings and events, and serve as a liaison among the partner institutions. This support structure will help maintain smooth operations and allow the researchers to focus on scientific and educational tasks. The Center will comply with all reporting requirements of the funding agency, including annual progress reports and financial reports, which will be prepared by the administrative team with input from the Executive Committee.

Overall, the management plan emphasizes clear leadership roles, frequent communication, accountability through defined milestones, and incorporation of external expert advice. This structure is designed to keep a large and diverse team aligned toward the common goals of the Center, enabling us to adapt proactively and maintain excellence in both research and training throughout the grant period.

7 Expected Impact

If successful, the Research Center for Quantum Control will have far-reaching impacts on science, technology, and society:

- **Scientific Impact:** The Center will generate new knowledge at the intersection of quantum physics and control engineering. We anticipate foundational advances such as general principles for controlling high-dimensional quantum systems, improved algorithms for pulse optimization, and insights into the limits of controllability in the presence of noise. These contributions will be disseminated through publications in high-impact journals and presentations at major conferences, establishing the Center as a world leader in quantum control research.
- **Technological Impact on Quantum Systems:** By elevating the performance of quantum gates and reducing errors through sophisticated control, the Center will help unlock more computational power from near-term quantum processors. We expect to demonstrate that quantum algorithms run with our control techniques achieve better accuracy or greater circuit depth than those using standard control, thus enabling experiments in quantum simulation or small-scale computation that were previously infeasible. In quantum metrology and sensing, our control-enabled enhancement of coherence and entanglement should translate to sensors (e.g., atomic clocks, magnetometers) with record-breaking precision and stability. Furthermore, the software and hardware tools developed (such as advanced pulse sequence libraries and custom control electronics) will be transferrable to industry, accelerating the development of commercial quantum technologies.
- **Educational and Workforce Impact:** The interdisciplinary training provided through the Center will produce a cohort of scientists and engineers proficient in both quantum physics and control technology. These individuals will be highly sought after in the growing quantum industry as well as in academia. We anticipate that many of our graduates will become leaders who drive innovations in quantum computing startups, contribute to national labs, or build new academic research programs,

thereby propagating the Center’s influence. The new curriculum and teaching materials created (text-books, laboratory manuals, etc.) will be adopted more broadly, raising the level of quantum engineering education nationwide.

- **Infrastructure and Community Impact:** The experimental testbeds and open-source control software established by the Center will serve as lasting infrastructure. Other researchers will be able to build upon our platforms and tools, whether by accessing our testbeds through collaboration or by replicating our setups at their institutions. By publishing detailed technical designs and sharing data, we aim to set new standards in reproducibility and openness for quantum experiments. The collaborative network fostered by the Center, linking multiple universities and industry partners, will strengthen the quantum research community and encourage joint efforts beyond the scope of this project.
- **Societal and Economic Impact:** As quantum technologies move toward practical application, improved control translates directly into benefits like more reliable quantum computers for solving hard problems in chemistry or materials science, and more precise quantum sensors for navigation or medical imaging. The Center’s work will contribute to these advancements, potentially leading to economic opportunities (new products, startups, intellectual property) and societal benefits (e.g., better drug discovery through quantum simulation, improved GPS-free navigation via quantum sensors). Through our outreach and engagement, we also aim to increase public understanding of quantum science and inspire future students to enter STEM fields, thereby broadening the societal impact of our work.

In conclusion, the establishment of the Research Center for Quantum Control will address a pivotal challenge in the quantum revolution: how to exert exquisite control over complex quantum systems. Success in this endeavor will accelerate the realization of practical quantum computers, more powerful quantum simulators for scientific discovery, and ultra-sensitive measurement devices. The Center’s collaborative, multi-platform approach ensures that advances in one area inform and elevate others, creating a sum greater than the parts. With its combination of cutting-edge research, shared infrastructure, and strong educational mission, the Center is poised to make enduring contributions that will shape the trajectory of quantum technology for years to come.

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